

Kang Xu, Ph.D.

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Publication: Google Scholar | LinkedIn | GitHub

EDUCATION

Nanjing University

Ph.D. in Condensed Matter Physics

Sep. 2012 – Aug. 2018

Nanjing, China

China University of Petroleum (East China)

B.S. in Applied Physics

Sep. 2008 – Jul. 2012

Qingdao, China

TECHNICAL SKILLS

Quantum Control & Sensing: ODMR, Rabi/Ramsey, dynamical decoupling (XY8/CPMG), correlation protocols, NMR, pulse calibration, noise characterization.

Experimental Hardware: laser optics, confocal microscopy, scanning probe microscopy, AOM/EOM modulation, microwave/RF delivery, timing/trigger synchronization, high-speed digitizers / NI-SCOPE / PulseBlaster / AWG.

Software: Python, MATLAB, LabVIEW, Git, instrument control (VISA/SCPI, drivers/APIs, GUI, quantum simulation (QuTiP/QuaCCAToo), CAD/EDA: Inventor, Altium.

Nanofabrication & Devices: 2D heterostructures, cleanroom processing, optical/electrical characterization.

PROFESSIONAL EXPERIENCE

City College of New York (CCNY)

Postdoctoral Researcher — Quantum Sensing

New York, NY

Oct. 2022 – Present

- Designed and engineered an NV-center quantum measurement platform integrating laser optics, microwave control, and synchronized control interfaces for nanoscale NMR and interfacial dynamics studies.
- Implemented pulse-sequence control with a multichannel AWG and PulseBlaster, enabling reliable execution of NMR protocols (e.g., XY8/CPMG-based sequences) with automated parameter sweeps.
- Developed magnetic-field calibration and measurement optimization routines (SNR/contrast, NV drift tracking), improving day-to-day reproducibility and stability.
- Led end-to-end troubleshooting across optics, RF/microwave chain, timing/DAQ, and software control, resolving integration issues and improving platform reliability in a multi-user lab.

Tsinghua University

Postdoctoral Researcher

Beijing, China

May. 2020 – Aug. 2022

- Set up and maintained a cryostat-based optoelectronic characterization station, performing optical alignment, automated wavelength sweeps, and modulation/lock-in readout of photoresponse—enabling repeatable electrical and optoelectronic measurements of semiconductor and 2D devices.
- Performed optical/electrical characterization of layered semiconductors and heterostructures, exploring material properties for next-generation optoelectronic devices.

Quantum Design China

Technical Sales Engineer (Scientific Instrumentation)

Shanghai, China

Oct. 2018 – Feb. 2020

- Installed and commissioned near-field optical microscope systems at customer sites; delivered end-user training and acceptance tests.
- Troubleshot hardware/software issues under time pressure, coordinating with R&D and application teams to close issues and reduce downtime.

Nanjing University

Ph.D. Researcher

Nanjing, China

Sep. 2012 – Aug. 2018

- Performed cleanroom nanofabrication and device testing for suspended graphene; built AFM-based measurement setups to quantify electromechanical transport of graphene.
- Designed in-situ strain tuning platforms in an Oxford cryostat for low-temperature transport measurements of low-dimensional materials.

Spin-Probe-Enabled Sensing of Fluids in Confined Geometries and Interfaces*NSF-DFG Confine Project, Co-investigator, \$600,000*

- G. García-Arellano, **K. Xu**, A. Ramanathan, J. Li, G. I. López-Morales, X. Roy, C. E. Dreyer, and C. A. Meriles¹, “Erbium Probes of Magnetic Order in a Layered van der Waals Material,” *arxiv* (2026).
- **K. Xu**, K. E. R. K. Wasa, R. Khan, G. López, A. Wolcott, D. Pagliero, N. Giovambattista, and C. A. Meriles, “Nanoscale Color Center Sensing of Adsorbed Water in Contact with Oil,” *Advanced Functional Materials* published online (2025).
- D. Pagliero, R. Khan, K. E. R. K. Wasa, A. Bhardwaj, **K. Xu**, A. Wolcott, R. Boya, G. López, N. Giovambattista, and C. A. Meriles, “Slow water in engineered nano-channels revealed by color-center-enabled sensing,” *Nano Letters* 25, 9960 (2025).
- **K. Xu**, D. Pagliero, G.I. Lopez Morales, J. Flick, A. Wolcott, and C. A. Meriles, “Photoinduced charge injection from shallow point defects in diamond into water,” *ACS Appl. Mater. Interfaces* 16, 37226–37233 (2024).

Optoelectronic Devices of Perovskite Semiconductor Microcavity Exciton-Polaritons*NSFC Project, Main contributor to the optoelectronic section, ¥2,700,000*

- X. Chen#, **K. Xu**#, T. Qin, Y. Wang, Y. Chen, H. Liu, and Q. Xiong, “Bulk photovoltaic effect in two-dimensional ferroelectric semiconductor α -In₂Se₃,” *Nanoscale* 9, (2025). (**Contributed equally**)
- X. Chen, **K. Xu**, X. Zhang, H. Liu, Q. Xiong, “Research progress of bulk photovoltaic effect in two-dimensional materials,” *Acta Phys. Sin.* 72(23): 237201 (2023).

Physical Property Modulation and Device Research of 2D Materials*National 973 Project, Main contributor to the strain field coupling section, ¥4,420,000*

- **K. Xu**#, K. Wang#, W. Zhao#, W. Bao, E. Liu, Y. Ren, M. Wang, Y. Fu, J. Zeng, Z. Li, W. Zhou, F. Song, X. Wang, Y. Shi, X. Wan, M. S. Fuhrer, B. Wang, Z. Qiao, F. Miao, D. Xing, “The positive piezoconductive effect in graphene,” *Nature Communications* 6, 8119 (2015). (**Contributed equally**)
- **K. Xu**, E. Liu, F. Miao, “Graphene induced hole conduction of MoS₂,” *China Science paper* 13, 5 (2018).
- Z. Ma, S. Yan, F. Chen, Y. Dai, Z. Liu, **K. Xu**, T. Xu, Z. Tong, M. Chen, L. Wang, P. Wang, L. Sun, B. Cheng, S. Liang, F. Miao, “Ultrasensitive mechanical sensor using tunable ordered array of metallic and insulating states in vanadium dioxide,” *Chinese Physics Letters*, 41, 077101 (2024).
- Y. Wang, C. Wang, S. Liang, Z. Ma, **K. Xu**, X. Liu, L. Zhang, A. Admasu, S. Cheong, L. Wang, M. Chen, Z. Liu, B. Cheng, W. Ji, F. Miao, “Strain-sensitive magnetization reversal of a van der Waals magnet,” *Advanced Materials* 32, 2004533 (2020).
- M. Wang, S. Cai, C. Pan, C. Wang, X. Lian, Y. Zhuo, **K. Xu**, T. Cao, X. Pan, B. Wang, S. Liang, J. Yang, P. Wang, F. Miao, “Robust memristors based on layered two-dimensional materials,” *Nature Electronics* 1, 130 (2018).
- J. Zeng, E. Liu, Y. Fu, Z. Chen, C. Pan, C. Wang, M. Wang, Y. Wang, **K. Xu**, S. Cai, X. Yan, Y. Wang, X. Liu, P. Wang, S. Liang, Y. Cui, H. Wang, H. Yuan, F. Miao, “Gate-Induced interfacial superconductivity in 1T-SnSe₂,” *Nano Letters* 18, 1410 (2018).
- Y. Fu, M. Long, A. Gao, Y. Wang, C. Pan, X. Liu, J. Zeng, **K. Xu**, L. Zhang, E. Liu, W. Hu, X. Wang, F. Miao, “Intrinsic p-type W-based transition metal dichalcogenide by substitutional Ta-doping,” *Applied Physics Letters* 111, 043502 (2017).
- Y. Wang, E. Liu, H. Liu, Y. Pan, L. Zhang, J. Zeng, Y. Fu, M. Wang, **K. Xu**, Z. Huang, Z. Wang, H. Lu, D. Xing, B. Wang, X. Wan, F. Miao, “Gate-tunable negative longitudinal magnetoresistance in the predicted type-II Weyl semimetal WTe₂,” *Nature Communications* 7, 13142 (2016).

CONFERENCE PRESENTATIONS

2026 MRS spring meeting	2025
<i>Poster</i>	<i>Hawaii, USA</i>
2025 APS March meeting	2025
<i>Speaker</i>	<i>CA, USA</i>
2024 APS March meeting	2024
<i>Speaker</i>	<i>Minneapolis, USA</i>
2016 APS March meeting	2016
<i>Speaker</i>	<i>Baltimore, USA</i>
International Symposium on 2D Materials	2016
<i>Poster</i>	<i>Shanghai, China</i>
Institute of Physics CAS National PhD Academic Forum	2016
<i>Poster</i>	<i>2 Beijing, China</i>

SCHOLARSHIPS & AWARDS

First prize winner of 4th Global NMR online conference <i>Global NMR</i>	2025
APS-Simons Travel and Professional Development Awards <i>Simons Foundation</i>	2025
Outstanding Graduates Honor Award <i>Nanjing University (10%)</i>	2018
Program A for Outstanding PhD Candidate <i>Nanjing University (1%)</i>	2017
National Scholarship for Graduate Students <i>Ministry of Education of the People's Republic of China (5%)</i>	2016
Principal Special Scholarship for Doctoral Students <i>Nanjing University (2%)</i>	2016
Oxford Instruments Rising Star China Scholarship <i>Oxford Instruments</i>	2016
First Prize Scholarship for Excellent Graduate Students <i>Collaborative Innovation Center of Advanced Microstructures (CICAM) (5%)</i>	2015
Award for Excellence in Research for Graduate Students <i>National Laboratory of Solid-State Microstructures (NLSSM) (10%)</i>	2015